

INGALLS FIELD, VA, USA

WMO: 724115

Lat: 37.950N

Lon: 79.833W

Elev: 3793

StdP: 12.79

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 93757

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	1.0	6.9	-9.3	3.8	10.5	-4.4	5.0	15.9	33.6	21.4	31.4	24.3	18.7	290	0.557

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	13.2	81.8	67.8	79.5	66.9	77.3	65.9	72.1	77.0	70.4	75.3	69.0	74.1	7.9	280

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
70.5	129.5	74.3	69.6	125.4	73.1	67.5	116.6	71.3	38.5	76.9	37.0	75.3	35.6	74.9	80.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
28.1	25.5	23.1	DB	-4.3	85.3	7.0	4.4	-9.4	88.4	-13.5	91.0	-17.4	93.5	-22.5	96.7	(4)
			WB	-4.1	73.5	6.3	2.7	-8.6	75.4	-12.3	77.0	-15.8	78.6	-20.4	80.6	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	49.7	28.3	31.5	38.5	49.2	57.9	65.3	69.1	68.0	61.7	51.7	40.5	33.1
	DBStd	17.16	12.89	11.86	11.55	10.35	8.92	6.07	4.85	5.51	7.01	9.02	10.61	11.33
	HDD50	2706	678	524	387	139	32	2	0	2	5	90	314	533
	HDD65	6018	1137	938	822	479	243	67	19	31	141	418	734	989
	CDD50	2586	6	6	30	115	278	462	592	561	357	142	29	8
	CDD65	424	0	0	0	5	24	76	146	124	43	6	0	0
	CDH74	1398	0	0	1	21	83	219	527	403	127	17	0	0
	CDH80	180	0	0	0	1	9	16	81	53	13	7	0	0
Wind	WSAvg	10.7	14.6	13.6	12.1	11.7	9.9	8.9	7.9	7.1	7.8	10.2	11.9	13.0
Precipitation	PrecAvg	39.6	3.2	2.6	3.8	3.2	3.9	3.8	3.9	3.0	3.7	2.4	3.0	3.1
	PrecMax	56.2	7.0	6.4	7.1	5.3	7.8	8.7	6.2	4.7	9.8	6.7	6.0	6.7
	PrecMin	33.2	1.0	0.7	1.2	1.0	0.8	0.9	1.7	0.9	0.4	0.0	0.4	1.6
	PrecStd	5.6	1.7	1.4	1.5	1.2	1.8	2.0	1.3	1.1	2.5	1.8	1.5	1.2
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	58.9	61.3	69.9	77.1	80.8	82.0	85.7	84.3	81.9	74.9	66.1	60.7
		MCWB	54.3	53.3	54.9	58.2	62.6	67.7	68.0	69.6	65.2	64.6	56.1	55.8
	2%	DB	54.8	55.8	63.5	72.2	75.5	79.3	81.9	81.3	78.1	70.5	61.8	55.5
		MCWB	51.3	50.8	52.2	56.2	62.3	67.1	68.1	69.0	64.8	60.6	52.1	51.9
	5%	DB	51.8	52.2	58.9	67.8	72.9	76.6	79.4	78.9	74.0	66.3	57.4	53.5
		MCWB	49.1	46.6	49.3	53.9	61.5	65.9	67.5	67.9	63.4	58.8	50.6	49.5
	10%	DB	46.3	48.0	54.8	63.7	70.2	73.5	76.8	75.2	71.8	63.8	54.9	49.5
		MCWB	42.9	43.1	47.2	52.8	60.4	64.9	66.5	66.4	62.6	57.0	49.1	45.8
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	55.7	56.7	58.0	62.6	68.2	71.9	74.6	73.7	71.2	68.1	59.2	57.6
		MCDB	58.4	59.5	65.4	70.1	73.3	76.8	79.3	79.5	75.1	72.6	63.2	58.9
	2%	WB	52.6	52.6	54.9	59.7	65.7	69.7	72.3	71.9	68.6	64.3	56.1	54.0
		MCDB	54.1	55.4	60.6	66.3	71.8	75.2	76.7	77.2	72.8	67.1	59.1	55.4
	5%	WB	48.7	48.0	52.2	57.3	63.9	68.2	70.4	70.2	66.5	61.2	53.7	50.3
		MCDB	50.5	50.7	57.0	63.4	69.6	74.0	75.2	75.5	70.6	65.1	56.3	52.3
	10%	WB	44.1	43.9	48.7	55.0	62.1	66.7	69.0	68.7	64.8	58.3	50.5	46.3
		MCDB	46.2	47.1	53.3	61.2	67.6	72.3	74.0	73.8	68.6	62.6	54.2	47.8
Mean Daily Temperature Range	5% DB	MDBR	14.0	15.0	16.0	17.0	15.8	14.1	13.2	13.5	13.3	13.9	13.9	12.9
		MCDBR	13.9	18.1	18.1	19.2	17.8	15.6	15.2	15.1	15.4	15.9	15.3	13.8
		MCWBR	11.6	14.4	11.5	9.9	9.5	8.6	7.8	7.5	8.0	9.5	11.3	11.6
	5% WB	MCDBR	14.1	17.5	16.7	16.2	15.8	13.8	13.1	13.4	12.9	13.8	14.0	14.0
		MCWBR	13.1	16.2	12.1	10.6	9.6	8.2	7.9	7.3	7.6	9.9	12.5	13.3
Clear-Sky Solar Irradiance	taub	0.270	0.284	0.307	0.342	0.373	0.402	0.423	0.425	0.342	0.314	0.284	0.269	
	taud	2.545	2.491	2.484	2.408	2.314	2.258	2.230	2.195	2.436	2.515	2.574	2.574	
	Ebn at Noon	295	303	304	297	288	278	271	268	288	287	287	288	
	Edh at Noon	23	29	32	36	41	43	44	44	33	28	23	22	
All-Sky Solar Radiation	RadAvg	665	905	1236	1566	1725	1895	1801	1646	1376	1067	779	584	
	RadStd	61	115	79	122	150	126	112	81	148	127	74	52	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only		N/A	N/A	N/A	-1.87	N/A	N/A	N/A	N/A	N/A	N/A
Regional (33 neighbors)		+0.63	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+227	+115

Nomenclature: See separate page